

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

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Finding reasoning paths :

- 1 . Consider the anatomy of the anatomical snuff box and the arteries involved in this region .
- 2 . Identify the anatomical structures that could be injured in a stab wound to the anatomical snuff box .
- 3 . Analyze the functional role of the Princeps pollicis artery and its implications in the context of bleeding from the snuff box .

Reasoning Process :

- 1 . **Anatomy of the Anatomical Snuff box**: The anatomical snuff box is an anatomical pocket located on the lateral side of the hand , bordered by the tendons of the abductor pollicis longus , extensor pollicis brevis , and extensor pollicis longus muscles . It is part of the thenar region of the hand .
- 2 . **Arteries in the Anatomical Snuff box**: The Princeps pollicis artery , which is the terminal branch of the deep palmar branch of the ulnar artery , runs within the anatomical snuff box . It is responsible for supplying blood to the thumb , particularly to the terminal phalanx and proximal interphalangeal joint .
- 3 . **Impact of Injury**: A penetrating stab wound to the anatomical snuff box could potentially damage the Princeps pollicis artery , leading to excessive bleeding . The deep palmar branch of the ulnar artery gives off the Princeps pollicis artery to supply the digital arteries and adjacent tissues .
- 4 . **Clinical Presentation**: The teenager presents with severe bleeding from the right hand , which aligns with the anatomical region involved . The Princeps pollicis artery is the most likely artery to be damaged due to its proximity and function within the anatomical snuff box .

Conclusion :

The damage is most likely to the Princeps pollicis artery , which is the terminal branch of the deep palmar branch of the ulnar artery and runs within the anatomical snuff box . This explains the excessive bleeding from the injured hand .

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< answer >

Princeps pollicis artery

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